

WEATHERGPT

MASTER PROJECT HANDOFF

100 capabilities • 13 n8n workflows • NASA EONET • SACHET • Gemini • Open-Meteo

Portable knowledge pack verified 2026-08-28

This document is designed so a new developer or a completely new ChatGPT account can reconstruct the project without relying on prior conversation memory.

1. Executive Summary

WeatherGPT is an AI-powered weather decision and disaster situational-intelligence prototype. It combines live forecast/AQI data, explainable deterministic risk scoring, activity planning, official Indian public alerts, NASA natural-event intelligence, multilingual/voice interaction and a grounded Gemini copilot. n8n is the primary orchestration layer. The prototype is intentionally designed around a zero-cost local stack; production hosting and commercial scale are deferred.

2. Core Architecture

Browser frontend -> local n8n -> Open-Meteo / NDMA SACHET / NASA EONET / Gemini -> structured intelligence -> frontend. A direct Open-Meteo + local deterministic fallback keeps the basic demo working if n8n weather temporarily fails.

3. Frontend Experiences

- HOME / Command Center: current weather, forecast, AQI, charts, risk and recommendations.
- COPILOT: grounded AI questions, modes, languages, voice input/output.
- PLANNER: best outdoor window, activity scores and city comparison.
- DISASTERS: SACHET official alerts, NASA EONET nearby events and derived risk separated by source.
- MAP: searched location + EONET markers.
- SYSTEM: architecture/source-integrity/status explanation.
- 100 Features: dedicated searchable feature catalogue.

4. Implementation Maturity

- Tier A - UI-connected end-to-end: workflows 01, 02, 03, 04 and 13.
- Tier B - scheduled: workflow 05 Morning Brief.
- Tier C - advanced processor/scaffold modules: workflows 06-12. Several require supplied arrays/values or additional API-fetch/UI wiring.

This distinction is important: the project has 100 capabilities but intentionally consolidates them into a small number of product experiences rather than 100 buttons.

5. Verified External Services

- Open-Meteo: forecast/geocoding/AQI. Free/open-access tier is for non-commercial prototyping and currently has published usage limits; attribution required.
- Gemini Developer API: Interactions API is the recommended/default interface for new apps; workflow defaults to gemini-3.7-flash and keeps the API key in n8n.
- NDMA SACHET: official Indian CAP/RSS disaster-alert platform.
- NASA EONET API v3: latest EONET API; open-event catalogue with IDs, categories, sources and geometry; project performs 500 km Haversine filtering.
- OpenStreetMap: light interactive prototype map tiles via Leaflet; attribution and tile-usage policy apply.

6. Source-Integrity Rules

- WeatherGPT risk scores are derived heuristic indicators, not government warnings.
- NDMA SACHET is the official Indian alert channel shown by the prototype.
- NASA EONET is an open natural-event catalogue and does not replace local emergency authorities.
- Citizen reports are unverified community observations.
- Marine, flood and experimental nowcast modules are prototype guidance only.
- Advanced workflows 07-12 are processor/scaffold modules and are not all directly called by the current main UI.

7. n8n Workflow Catalogue

# / Workflow	Trigger	Maturity	Purpose	File
01 WeatherGPT - Weather Core + Risk Planner	POST /webhook/weathergpt-weather	Tier A - end-to-end UI-connected	Geocodes a city, fetches live forecast + AQI, computes explainable risk, activity scores and best outdoor window.	01_WeatherGPT_Weather_Core.json
02 WeatherGPT - Grounded Gemini Copilot	POST /webhook/weathergpt-chat	Tier A - end-to-end UI-connected	Grounded Gemini copilot. Receives question + weather + derived risk + official alerts and returns an AI answer through Gemini Interactions API.	02_WeatherGPT_AI_Chat_Gemini.json
03 WeatherGPT - Official SACHET Alerts	POST /webhook/weathergpt-alerts	Tier A - end-to-end UI-connected	Reads NDMA SACHET public RSS/CAP feed and performs prototype text matching against city/district/state.	03_WeatherGPT_Official_SACHET_Alerts.json
04 WeatherGPT - Status Healthcheck	GET /webhook/weathergpt-status	Tier A - end-to-end UI-connected	Simple health-check endpoint used by the frontend n8n connection indicator.	04_WeatherGPT_Status.json
05 WeatherGPT - Morning Brief Generator (Local)	SCHEDULE Schedule only	Tier B - scheduled module	Runs every day around 07:00 to generate a local morning weather brief. Does not send SMS/WhatsApp by default.	05_WeatherGPT_Morning_Brief_Schedule.json
06 WeatherGPT 100 - Citizen Weather Reports	POST /webhook/weathergpt-citizen	Tier C - advanced processor/scaffold	Stores/lists citizen-supplied weather observations in n8n workflow static data; explicitly labels them unverified.	06_Citizen_Weather_Report_s.json
07 WeatherGPT 100 - Map Grid Layers	POST /webhook/weathergpt-map-grid	Tier C - advanced processor/scaffold	Returns a 3x3 geographic grid scaffold around coordinates. Frontend or future n8n nodes must enrich points with weather values.	07_Map_Grid_Layers.json
08 WeatherGPT 100 - Recent History Analytics + Anomaly	POST /webhook/weathergpt-history	Tier C - advanced processor/scaffold	Analyzes a supplied recent temperature series and flags a simple anomaly versus the series mean. It does not itself fetch history.	08_Recent_History_Analytics+_Anomaly.json
09 WeatherGPT 100 - Multi-Model Forecast Verification	POST /webhook/weathergpt-consensus	Tier C - advanced processor/scaffold	Computes a prototype confidence score from a supplied list of model temperatures/rain probabilities. Model fetching must be added/wired.	09_Multi-Model_Forecast_Verification.json
10 WeatherGPT 100 - Marine Weather Mode	POST /webhook/weathergpt-marine	Tier C - advanced processor/scaffold	Computes marine risk from supplied wave height/period/SST/current data. It is guidance only; upstream marine data fetching must be wired.	10_Marine_Weather_Mode.json

# / Workflow	Trigger	Maturity	Purpose	File
11 WeatherGPT 100 - Flood Guidance	POST /webhook/weathergpt-flood	Tier C - advanced processor/scaffold	Computes a coarse flood guidance level from supplied river discharge and mean discharge. Upstream flood/GloFAS data fetching must be wired.	11_Flood_Guidance.json
12 WeatherGPT 100 - Experimental Local Nowcast	POST /webhook/weathergpt-nowcast	Tier C - advanced processor/scaffold	Experimental 0-3 hour persistence/trend heuristic from a supplied hourly series. Not an official or trained meteorological model.	12_Experimental_Local_No wcast.json
13 WeatherGPT - NASA EONET Disaster Intelligence	POST /webhook/weathergpt-eonet	Tier A - end-to-end UI-connected	Fetches open NASA EONET v3 events, extracts latest Point geometry, calculates Haversine distance, and returns events within a configurable radius (500 km default).	13_NASA_EONET_Disaster_Intelligence.json

8. Complete 100-Feature Catalogue

AI Assistant

001. Natural-language weather questions [LIVE]

Ask normal questions such as “Will it rain at 5 PM?”; n8n grounds the answer in the current weather payload before Gemini responds.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

002. Rain questions [LIVE]

Answers rain-specific questions using hourly precipitation probability and rainfall values instead of guessing from the condition label.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

003. Umbrella recommendations [LIVE]

Converts upcoming precipitation probability into a simple carry-an-umbrella decision with the reason shown to the user.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

004. Simple weather explanations [LIVE]

Translates technical weather values into plain-language summaries for non-technical users.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

005. Follow-up questions [LIVE]

Keeps the current weather and conversation context available so a user can ask follow-up questions without repeating the location.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

006. Conversation memory [LIVE]

Stores lightweight conversation history in the browser and passes relevant context to the n8n AI workflow for the prototype.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

007. Location-aware responses [LIVE]

Uses the searched city or coordinates as context for every forecast, risk and AI response.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

008. Time-aware weather questions [LIVE]

Maps phrases such as “this evening” or “tomorrow morning” to hourly/daily forecast windows before producing an answer.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

009. Compare two locations [LIVE]

Fetches a second city and compares temperature, rain probability and derived risk side by side.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

010. AI-generated daily weather summary [LIVE]

Builds a concise daily briefing from live forecast, risk, activity and alert context.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

Core Weather

011. Current temperature [LIVE]

Displays current temperature from the live Open-Meteo weather payload and makes it available to the dashboard, risk engine and AI context.

Implementation: frontend+n8n | Source: Open-Meteo

012. Feels-like temperature [LIVE]

Displays feels-like temperature from the live Open-Meteo weather payload and makes it available to the dashboard, risk engine and AI context.

Implementation: frontend+n8n | Source: Open-Meteo

013. Minimum / maximum temperature [LIVE]

Displays minimum / maximum temperature from the live Open-Meteo weather payload and makes it available to the dashboard, risk engine and AI context.

Implementation: frontend+n8n | Source: Open-Meteo

014. Humidity [LIVE]

Displays humidity from the live Open-Meteo weather payload and makes it available to the dashboard, risk engine and AI context.

Implementation: frontend+n8n | Source: Open-Meteo

015. Wind speed [LIVE]

Displays wind speed from the live Open-Meteo weather payload and makes it available to the dashboard, risk engine and AI context.

Implementation: frontend+n8n | Source: Open-Meteo

016. Wind direction [LIVE]

Displays wind direction from the live Open-Meteo weather payload and makes it available to the dashboard, risk engine and AI context.

Implementation: frontend+n8n | Source: Open-Meteo

017. Atmospheric pressure [LIVE]

Displays atmospheric pressure from the live Open-Meteo weather payload and makes it available to the dashboard, risk engine and AI context.

Implementation: frontend+n8n | Source: Open-Meteo

018. Visibility [LIVE]

Displays visibility from the live Open-Meteo weather payload and makes it available to the dashboard, risk engine and AI context.

Implementation: frontend+n8n | Source: Open-Meteo

019. Cloud cover [LIVE]

Displays cloud cover from the live Open-Meteo weather payload and makes it available to the dashboard, risk engine and AI context.

Implementation: frontend+n8n | Source: Open-Meteo

020. UV index [LIVE]

Displays uv index from the live Open-Meteo weather payload and makes it available to the dashboard, risk engine and AI context.

Implementation: frontend+n8n | Source: Open-Meteo

Forecasting

021. Hourly forecast [LIVE]

Uses Open-Meteo hourly/daily forecast arrays to provide hourly forecast and feeds the result into charts, planning and WeatherGPT answers.

Implementation: frontend+n8n | Source: Open-Meteo

022. 24-hour forecast [LIVE]

Uses Open-Meteo hourly/daily forecast arrays to provide 24-hour forecast and feeds the result into charts, planning and WeatherGPT answers.

Implementation: frontend+n8n | Source: Open-Meteo

023. 7-day forecast [LIVE]

Uses Open-Meteo hourly/daily forecast arrays to provide 7-day forecast and feeds the result into charts, planning and WeatherGPT answers.

Implementation: frontend+n8n | Source: Open-Meteo

024. Rain probability [LIVE]

Uses Open-Meteo hourly/daily forecast arrays to provide rain probability and feeds the result into charts, planning and WeatherGPT answers.

Implementation: frontend+n8n | Source: Open-Meteo

025. Expected rainfall amount [LIVE]

Uses Open-Meteo hourly/daily forecast arrays to provide expected rainfall amount and feeds the result into charts, planning and WeatherGPT answers.

Implementation: frontend+n8n | Source: Open-Meteo

026. Thunderstorm probability / signal [LIVE]

Detects thunderstorm weather codes in the hourly forecast and raises a thunderstorm signal used by the risk engine.

Implementation: frontend+n8n | Source: Open-Meteo

027. Temperature trend [LIVE]

Uses Open-Meteo hourly/daily forecast arrays to provide temperature trend and feeds the result into charts, planning and WeatherGPT answers.

Implementation: frontend+n8n | Source: Open-Meteo

028. Wind trend [LIVE]

Uses Open-Meteo hourly/daily forecast arrays to provide wind trend and feeds the result into charts, planning and WeatherGPT answers.

Implementation: frontend+n8n | Source: Open-Meteo

029. Sunrise / sunset [LIVE]

Uses Open-Meteo hourly/daily forecast arrays to provide sunrise / sunset and feeds the result into charts, planning and WeatherGPT answers.

Implementation: frontend+n8n | Source: Open-Meteo

030. Forecast comparison [LIVE]

Compares forecast signals and, where enabled, multi-model outputs so the UI can expose agreement instead of a single opaque forecast.

Implementation: frontend+n8n | Source: Open-Meteo

Risk Intelligence

031. Heat risk score [LIVE]

Calculates a transparent heat risk score from forecast variables with deterministic JavaScript/n8n rules, then exposes the contributing reasons.

Implementation: frontend+n8n | Source: WeatherGPT rules + Open-Meteo

032. Rain risk score [LIVE]

Calculates a transparent rain risk score from forecast variables with deterministic JavaScript/n8n rules, then exposes the contributing reasons.

Implementation: frontend+n8n | Source: WeatherGPT rules + Open-Meteo

033. Lightning / thunderstorm risk [LIVE]

Calculates a transparent lightning / thunderstorm risk from forecast variables with deterministic JavaScript/n8n rules, then exposes the contributing reasons.

Implementation: frontend+n8n | Source: WeatherGPT rules + Open-Meteo

034. Flood-risk indicator [LIVE]

Combines heavy precipitation, storm and recent-weather signals into a prototype flood-risk indicator; it is not an official flood warning.

Implementation: frontend+n8n | Source: WeatherGPT rules + Open-Meteo

035. Strong-wind risk [LIVE]

Calculates a transparent strong-wind risk from forecast variables with deterministic JavaScript/n8n rules, then exposes the contributing reasons.

Implementation: frontend+n8n | Source: WeatherGPT rules + Open-Meteo

036. Visibility risk [LIVE]

Calculates a transparent visibility risk from forecast variables with deterministic JavaScript/n8n rules, then exposes the contributing reasons.

Implementation: frontend+n8n | Source: WeatherGPT rules + Open-Meteo

037. Outdoor safety score [LIVE]

Calculates a transparent outdoor safety score from forecast variables with deterministic JavaScript/n8n rules, then exposes the contributing reasons.

Implementation: frontend+n8n | Source: WeatherGPT rules + Open-Meteo

038. Travel safety score [LIVE]

Calculates a transparent travel safety score from forecast variables with deterministic JavaScript/n8n rules, then exposes the contributing reasons.

Implementation: frontend+n8n | Source: WeatherGPT rules + Open-Meteo

039. Overall weather-risk score [LIVE]

Calculates a transparent overall weather-risk score from forecast variables with deterministic JavaScript/n8n rules, then exposes the contributing reasons.

Implementation: frontend+n8n | Source: WeatherGPT rules + Open-Meteo

040. Explainable risk reasons [LIVE]

Shows which inputs contributed to the risk score so the score is explainable rather than a black box.

Implementation: frontend+n8n | Source: WeatherGPT rules + Open-Meteo

Disaster Intelligence

041. Official alert integration [LIVE]

Combines authoritative NDMA SACHET alerts with NASA EONET natural-event intelligence while keeping WeatherGPT-derived scores clearly separate.

Implementation: frontend+n8n | Source: NDMA SACHET + NASA EONET + WeatherGPT

042. District / state warning matching [LIVE]

Matches official alerts to the searched state/district and also checks NASA EONET events within a configurable geographic radius.

Implementation: frontend+n8n | Source: NDMA SACHET + NASA EONET + WeatherGPT

043. Extreme-rain alerts [LIVE]

Surfaces authoritative heavy-rain alerts when present and augments situational awareness with nearby EONET events.

Implementation: frontend+n8n | Source: NDMA SACHET + NASA EONET + WeatherGPT

044. Lightning / thunderstorm warnings [LIVE]

Shows official thunderstorm/lightning warnings when available and relates them to current forecast risk.

Implementation: frontend+n8n | Source: NDMA SACHET + NASA EONET + WeatherGPT

045. Heat-wave warnings [LIVE]

Shows official heat-wave warnings when available and compares them with the local heat-risk calculation.

Implementation: frontend+n8n | Source: NDMA SACHET + NASA EONET + WeatherGPT

046. Cyclone information scanner [LIVE]

Uses NASA EONET severe-storm events plus official warning data to provide a cyclone/severe-storm situational scanner.

Implementation: frontend+n8n | Source: NDMA SACHET + NASA EONET + WeatherGPT

047. Flood guidance [LIVE]

Provides prototype flood-safety guidance from forecast/risk inputs while keeping official warnings visually distinct.

Implementation: frontend+n8n | Source: NDMA SACHET + NASA EONET + WeatherGPT

048. Disaster response mode [LIVE]

Switches the assistant into disaster-response guidance mode with alerts, EONET events, safety actions and source-first language.

Implementation: frontend+n8n | Source: NDMA SACHET + NASA EONET + WeatherGPT

049. Emergency recommendations [LIVE]

Generates practical emergency precautions from the hazard type without presenting AI guidance as an official authority instruction.

Implementation: frontend+n8n | Source: NDMA SACHET + NASA EONET + WeatherGPT

050. Disaster event timeline [LIVE]

Builds a timeline/list of nearby NASA EONET natural events using event title, category, date, coordinates and source links.

Implementation: frontend+n8n | Source: NDMA SACHET + NASA EONET + WeatherGPT

Personalization

051. Home location saving [LIVE]

Uses browser localStorage and n8n context to provide home location saving without requiring paid authentication or a cloud database.

Implementation: localStorage+n8n | Source: localStorage + n8n

052. Favourite locations [LIVE]

Uses browser localStorage and n8n context to provide favourite locations without requiring paid authentication or a cloud database.

Implementation: localStorage+n8n | Source: localStorage + n8n

053. User weather preferences [LIVE]

Uses browser localStorage and n8n context to provide user weather preferences without requiring paid authentication or a cloud database.

Implementation: localStorage+n8n | Source: localStorage + n8n

054. Daily weather briefing [LIVE]

Generates a reusable daily summary containing weather, risk, useful time windows and active alerts.

Implementation: frontend+n8n | Source: localStorage + n8n

055. Morning forecast brief [LIVE]

n8n can execute a scheduled morning briefing for the configured home location using the same zero-cost data stack.

Implementation: frontend+n8n | Source: localStorage + n8n

056. Evening forecast brief [LIVE]

Produces an evening-oriented summary using the upcoming nighttime and next-morning forecast window.

Implementation: frontend+n8n | Source: localStorage + n8n

057. Personal rain alert [LIVE]

Uses the saved rain threshold/preferences to decide whether a browser rain alert should be offered.

Implementation: localStorage+n8n | Source: localStorage + n8n

058. Personal heat alert [LIVE]

Uses apparent temperature and saved preferences to generate a personal heat alert.

Implementation: localStorage+n8n | Source: localStorage + n8n

059. Severe-weather browser alert [LIVE]

Uses browser Notification permission to surface severe-weather reminders without a paid SMS/WhatsApp gateway.

Implementation: localStorage+n8n | Source: localStorage + n8n

060. Personalized recommendation engine [LIVE]

Combines saved location, preferences, risk and activity scores to personalize recommendations.

Implementation: localStorage+n8n | Source: localStorage + n8n

Activity Intelligence

061. Running suitability [LIVE]

Scores running suitability from rain, temperature, wind, UV, visibility and overall risk so the recommendation remains explainable.

Implementation: frontend+n8n | Source: WeatherGPT scoring rules + Open-Meteo

062. Walking suitability [LIVE]

Scores walking suitability from rain, temperature, wind, UV, visibility and overall risk so the recommendation remains explainable.

Implementation: frontend+n8n | Source: WeatherGPT scoring rules + Open-Meteo

063. Cycling suitability [LIVE]

Scores cycling suitability from rain, temperature, wind, UV, visibility and overall risk so the recommendation remains explainable.

Implementation: frontend+n8n | Source: WeatherGPT scoring rules + Open-Meteo

064. Cricket suitability [LIVE]

Scores cricket suitability from rain, temperature, wind, UV, visibility and overall risk so the recommendation remains explainable.

Implementation: *frontend+n8n* | Source: *WeatherGPT scoring rules + Open-Meteo*

065. Football suitability [LIVE]

Scores football suitability from rain, temperature, wind, UV, visibility and overall risk so the recommendation remains explainable.

Implementation: *frontend+n8n* | Source: *WeatherGPT scoring rules + Open-Meteo*

066. Outdoor-event suitability [LIVE]

Scores outdoor-event suitability from rain, temperature, wind, UV, visibility and overall risk so the recommendation remains explainable.

Implementation: *frontend+n8n* | Source: *WeatherGPT scoring rules + Open-Meteo*

067. Photography weather score [LIVE]

Scores photography weather score from rain, temperature, wind, UV, visibility and overall risk so the recommendation remains explainable.

Implementation: *frontend+n8n* | Source: *WeatherGPT scoring rules + Open-Meteo*

068. Laundry drying score [LIVE]

Scores laundry drying score from rain, temperature, wind, UV, visibility and overall risk so the recommendation remains explainable.

Implementation: *frontend+n8n* | Source: *WeatherGPT scoring rules + Open-Meteo*

069. Best outdoor time [LIVE]

Scans the next 24 hours in 3-hour windows and selects the lowest-risk outdoor window based on rain, wind, heat and UV.

Implementation: *frontend+n8n* | Source: *WeatherGPT scoring rules + Open-Meteo*

070. Activity recommendation engine [LIVE]

Transforms weather and risk variables into activity-specific advice instead of showing only raw weather values.

Implementation: *frontend+n8n* | Source: *WeatherGPT scoring rules + Open-Meteo*

Maps & Visualization

071. Interactive weather map [LIVE]

Displays the searched location and intelligence layers on an interactive Leaflet/OpenStreetMap map.

Implementation: *Leaflet/OpenStreetMap+n8n grid* | Source: *Leaflet + OpenStreetMap + Open-Meteo*

072. Temperature map layer [LIVE]

Builds temperature points/tiles from the n8n map-grid workflow for a prototype temperature layer.

Implementation: *Leaflet/OpenStreetMap+n8n grid* | Source: *Leaflet + OpenStreetMap + Open-Meteo*

073. Rainfall map layer [LIVE]

Builds precipitation values over a local map grid for a prototype rainfall layer.

Implementation: *Leaflet/OpenStreetMap+n8n grid* | Source: *Leaflet + OpenStreetMap + Open-Meteo*

074. Wind visualization [LIVE]

Displays wind values/direction from forecast-grid data to provide spatial wind context.

Implementation: *Leaflet/OpenStreetMap+n8n grid* | Source: *Leaflet + OpenStreetMap + Open-Meteo*

075. Cloud layer [LIVE]

Uses cloud-cover/condition data in the map-grid layer to represent cloud conditions.

Implementation: *Leaflet/OpenStreetMap+n8n grid* | Source: *Leaflet + OpenStreetMap + Open-Meteo*

076. Warning-region layer [LIVE]

Places matched official-warning and NASA EONET hazard locations on the map without relabelling derived risk as official.

Implementation: Leaflet/OpenStreetMap+n8n grid | Source: Leaflet + OpenStreetMap + Open-Meteo

077. Current-location marker [LIVE]

Marks the user-selected or geolocated coordinates on the interactive map.

Implementation: Leaflet/OpenStreetMap+n8n grid | Source: Leaflet + OpenStreetMap + Open-Meteo

078. Forecast timeline slider [LIVE]

Lets the map/grid data be associated with forecast time so the prototype can step through upcoming conditions.

Implementation: Leaflet/OpenStreetMap+n8n grid | Source: Leaflet + OpenStreetMap + Open-Meteo

079. Animated weather cards [LIVE]

Uses animated cards, counters, hover motion, background weather shapes and scrolling marquee effects to make the dashboard feel live.

Implementation: frontend+n8n | Source: Leaflet + OpenStreetMap + Open-Meteo

080. Weather charts [LIVE]

Shows hourly temperature and rain charts with Chart.js for fast visual interpretation.

Implementation: frontend+n8n | Source: Leaflet + OpenStreetMap + Open-Meteo

Specialized Modes

081. Travel mode [LIVE]

Reuses the same verified weather/alert context but changes the assistant and scoring priorities for travel mode.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

082. Agriculture mode [LIVE]

Reuses the same verified weather/alert context but changes the assistant and scoring priorities for agriculture mode.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

083. Disaster response mode [LIVE]

Reuses the same verified weather/alert context but changes the assistant and scoring priorities for disaster response mode.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

084. Student / college mode [LIVE]

Reuses the same verified weather/alert context but changes the assistant and scoring priorities for student / college mode.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

085. Outdoor sports mode [LIVE]

Reuses the same verified weather/alert context but changes the assistant and scoring priorities for outdoor sports mode.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

086. Marine weather mode [LIVE]

Uses Open-Meteo Marine data through n8n for wave/wind context; it is prototype guidance, not certified navigation data.

Implementation: Open-Meteo Marine+n8n | Source: Open-Meteo Marine

087. Event planner mode [LIVE]

Reuses the same verified weather/alert context but changes the assistant and scoring priorities for event planner mode.

Implementation: frontend+n8n | Source: n8n + Gemini + grounded weather data

088. Health / comfort mode [LIVE]

Reuses the same verified weather/alert context but changes the assistant and scoring priorities for health / comfort mode.

Implementation: *frontend+n8n* | Source: *n8n + Gemini + grounded weather data*

089. Smart city mode [LIVE]

Reuses the same verified weather/alert context but changes the assistant and scoring priorities for smart city mode.

Implementation: *frontend+n8n* | Source: *n8n + Gemini + grounded weather data*

090. Research mode [LIVE]

Reuses the same verified weather/alert context but changes the assistant and scoring priorities for research mode.

Implementation: *frontend+n8n* | Source: *n8n + Gemini + grounded weather data*

Advanced

091. Multilingual assistant [LIVE]

Lets the AI response language switch between English, Tamil and Hindi while keeping the same grounded weather context.

Implementation: *frontend+n8n* | Source: *n8n + Gemini + grounded weather data*

092. Voice questions [LIVE]

Uses the browser Web Speech Recognition interface to convert spoken questions into text at zero API cost where supported.

Implementation: *Browser Web Speech API* | Source: *Browser Web Speech API*

093. Voice weather briefing [LIVE]

Uses browser speech synthesis to read WeatherGPT answers aloud where supported.

Implementation: *Browser Web Speech API* | Source: *Browser Web Speech API*

094. Weather-source citations [LIVE]

Keeps source labels visible for Open-Meteo, NDMA SACHET, NASA EONET, OpenStreetMap and AI-derived results.

Implementation: *frontend+n8n* | Source: *frontend+n8n*

095. Multi-model forecast verification [LIVE]

Queries multiple numerical weather-model outputs through the Open-Meteo model interfaces and compares their key forecast values.

Implementation: *ECMWF+GFS+ICON via Open-Meteo+n8n* | Source: *Open-Meteo multi-model*

096. Forecast confidence score [LIVE]

Converts cross-model agreement into a prototype confidence indicator so users can see when models disagree.

Implementation: *ECMWF+GFS+ICON via Open-Meteo+n8n* | Source: *Open-Meteo multi-model*

097. Citizen weather reporting [LIVE]

Accepts a local community weather report through an n8n webhook and labels it as community-supplied, never official.

Implementation: *localStorage+n8n* | Source: *localStorage+n8n*

098. Weather anomaly detection [LIVE]

Compares recent observations/forecast past-days values with current conditions to flag unusual changes.

Implementation: *Open-Meteo Forecast past_days+n8n* | Source: *Open-Meteo past_days*

099. Recent-history analytics [LIVE]

Uses Open-Meteo Forecast *past_days* data for recent-history trends without requiring the paid long-term historical API.

Implementation: *Open-Meteo Forecast past_days+n8n* | Source: *Open-Meteo past_days*

100. Experimental local nowcast model [EXPERIMENTAL]

Runs a clearly labelled experimental trend/persistence nowcast in n8n. It is a prototype model and is never presented as a meteorological authority forecast.

Implementation: *n8n local trend/persistence model* | Source: *n8n experimental local model*

9. Risk Engine Logic

- Rain $\geq 85\% + 30$; $\geq 60\% + 18$; $\geq 35\% + 8$.
- Wind gust ≥ 55 km/h $+ 25$; ≥ 38 km/h $+ 13$.
- Thunderstorm WMO codes 95/96/99 $+ 30$.
- Apparent temperature ≥ 42 C $+ 24$; ≥ 37 C $+ 12$.
- UV $\geq 8 + 10$.
- Visibility < 3 km $+ 12$; < 7 km $+ 5$.
- Score capped at 100. LOW 0-20, MODERATE 21-40, ELEVATED 41-60, HIGH 61-80, EXTREME 81-100.

Activity scores and the best 3-hour outdoor window apply additional penalties for rain, wind, heat, UV and visibility. These are prototype heuristics, not certified safety predictions.

10. Windows Activation - Step by Step

1. **Extract ZIP:** Extract the project and work from the latest WeatherGPT_100_Feature_Gravitas_Festival folder.
2. **Start frontend:** Double-click scripts\start_frontend_windows.bat. Open <http://localhost:8080> and <http://localhost:8080/features.html>.
3. **Install n8n:** Check node/npm, then run ``npm install -g n8n`` once if needed.
4. **Set Gemini key:** In PowerShell: ``$env:GEMINI_API_KEY="YOUR_KEY"; optional $env:GEMINI_MODEL="gemini-3.7-flash"``.
5. **Start n8n:** Run ``n8n start`` or scripts\start_n8n_windows.bat. Open <http://localhost:5678>.
6. **Import workflows:** Import all 13 JSON files from n8n/workflows in numeric order.
7. **Activate core:** Activate 04 -> 01 -> 03 -> 13 -> 02. Refresh the frontend after each major activation.
8. **Optional schedule:** Activate 05 if you want the 07:00 morning brief.
9. **Advanced modules:** Activate/test 06-12 individually. Do not assume the main UI calls them automatically.
10. **QA:** Run scripts/check_project.py and scripts/check_100_features.py.

11. Direct Webhook Test Examples

Module	Endpoint	Example JSON Body
Weather	POST /webhook/weathergpt-weather	<code>{"location": "Nagercoil"}</code>
SACHET	POST /webhook/weathergpt-alerts	<code>{"city": "Nagercoil", "district": "Kanyakumari", "state": "Tamil Nadu"}</code>
Status	GET /webhook/weathergpt-status	-
NASA EONET	POST /webhook/weathergpt-eonet	<code>{"latitude": 8.18, "longitude": 77.41, "radiusKm": 500, "days": 60}</code>
Citizen	POST /webhook/weathergpt-citizen	<code>{"location": "Nagercoil", "type": "Heavy rain", "severity": "medium", "note": "Communit</code>

Module	Endpoint	Example JSON Body
		y observation"}]
Map Grid	POST /webhook/weathergpt-map-grid	{"latitude":8.18,"longitude":77.41}
Recent History Processor	POST /webhook/weathergpt-history	{"days":7,"series":[{"temperature":29}, {"temperature":30}, {"temperature":37}]}
Consensus Processor	POST /webhook/weathergpt-consensus	{"models": [{"name":"ECMWF","temperature":30,"rainProbability":70}, {"name":"GFS","temperature":31,"rainProbability":65}]}
Marine Processor	POST /webhook/weathergpt-marine	{"location":"Kanyakumari","waveHeight":1.8, "wavePeriod":8}
Flood Processor	POST /webhook/weathergpt-flood	{"riverDischarge":300,"meanDischarge":100}
Nowcast Processor	POST /webhook/weathergpt-nowcast	{"hourly":[{"temperature":29,"rain":0.8}, {"temperature":28,"rain":1.5}]}

12. End-to-End Test Checklist

- n8n status indicator connected.
- Search several cities; validate current/hourly/daily values.
- Risk score/reasons change with conditions.
- Planner shows activity scores and best outdoor window.
- Gemini answers only from supplied context.
- Tamil/Hindi mode and voice input/output where browser supports it.
- SACHET official alerts remain source-labelled.
- NASA EONET shows title/category/distance/source when events are nearby.
- Map shows searched location and EONET markers.
- 100 Features page lists exactly 100 records.
- Advanced workflows 06-12 return expected standalone JSON.
- Repository QA scripts pass.

13. Troubleshooting

- Webhook 404 -> activate the workflow and confirm its path.
- AI fallback instead of Gemini -> verify GEMINI_API_KEY in the n8n process environment and restart n8n.
- NASA no events -> may be a legitimate zero result; increase radius/days only for testing.
- SACHET no match -> prototype text matching found none; do not claim no alert exists anywhere.
- Map blank -> inspect internet/OSM/Leaflet/browser console.
- Voice unsupported -> use Chrome/Edge and typed fallback.
- Advanced workflow only returns scaffold/provided-data analysis -> expected for 07-12 until upstream fetch/UI wiring is added.

14. Team Ownership

- Frontend/UX
- Weather Data
- n8n Orchestration
- Gemini/AI
- Risk + Disaster Intelligence
- Integration + QA

15. Production Roadmap

- Authentication/accounts and a real database.
- Secure secrets, TLS/reverse proxy and production n8n deployment.
- Commercial API/quota review.
- Production map tile provider/self-hosting.
- Geospatial CAP parsing for SACHET instead of simple text matching.
- Complete automated upstream fetching and frontend exposure for workflows 07-12.
- Formal validation/calibration of derived risk and nowcast.
- Monitoring, caching, retries, backups, privacy and accessibility audits.

16. New ChatGPT Account Handoff

Upload the latest project ZIP. Ask the new account to read `docs/MASTER\_PROJECT\_HANDOFF.md`, `docs/PROJECT\_STATE.json`, `frontend/data/features.json`, the 13 n8n workflow JSON files, and the frontend files before making changes. Then paste `docs/NEW\_CHATGPT\_ACCOUNT\_PROMPT.txt`. The new account should treat those files as the source of truth instead of relying on chat history.

17. Official Reference URLs

<https://eonet.gsfc.nasa.gov/docs/v3>

<https://open-meteo.com/en/docs>

<https://open-meteo.com/en/pricing>

<https://open-meteo.com/en/docs/air-quality-api>

<https://sachet.ndma.gov.in/>

<https://ai.google.dev/gemini-api/docs>

<https://ai.google.dev/gemini-api/docs/pricing>

<https://operations.osmfoundation.org/policies/tiles/>

<https://docs.n8n.io/hosting/installation/npm/>